2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.365 Advanced Networking

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a

i. The Lancaster University Campus Network is a complex configuration of switches and routers, providing connectivity to many thousands of devices and end users. Explain the advantages and disadvantages of introducing a Quality of Service architecture such as Integrated Service and RSVP into the campus network.

[4 marks]

ii. Beyond the campus, networks such as JANET and GEANT scale upwards to support many millions of users across the UK and Europe. Describe the advantages and disadvantages of using Differentiated Services in such an environment.

[4 marks]

1.b At the highest level of abstraction, a router is divided into 4 key components, input ports, output ports, a routing processor, and a switching fabric. Describe in detail the main activities carried out by the input ports and output ports.

[4 marks]

1.c Explain the difference between a "maximum guaranteed bandwidth" and a "minimum guaranteed bandwidth".

[2 marks]

1.d Outline the basic principles of the Random Early Detection algorithm.

[3 marks]

1.e A layer-N flow is a sequence of packets with one common layer-N characteristic. Use an example to describe how a flow can be identified at layer 3. Repeat the task for a flow at layer 4.

[4 marks]

1.f Traffic control techniques exist that operate at a variety of different time scales. Briefly outline a technique that can be used in the context of each of the following timescales:

i. Day or longer
ii. Per call or session
iii. Burst
iv. Per packet

[4 marks]

Total 25 marks

Question 2

2.a Considering the Integrated Services and RSVP model of Quality of Service support:

i. Explain what is meant by "flow based soft state". **[2 marks]**

ii. Briefly outline the three levels of service proposed within the Integrated Services model. **[3 marks]**

iii. Describe the two primary messages used by RSVP. **[2 marks]**

2.b The two main DiffServ Per-Hop Behaviours are known as Expedited Forwarding (EF) and Assured Forwarding (AF).
Give your opinion as to which of the two is easier to implement, and your reasoning behind this. You should answer firstly from the perspective of a router vendor, and secondly from the perspective of a service provider.

[4 marks]

2.c In order to provide an effective Differentiated Services offering, boundary routers are likely to take part in both *admission control*, *packet marking*, and *traffic shaping*. Briefly describe each of these activities.

[3 marks]

2.d You have been commissioned by a mobile phone network provider to look at the potential impact on their network of multihomed devices, devices that have multiple network interfaces, capable of connecting to more than one network simultaneously.

i. What are the benefits of multihoming? **[1 mark]**

ii. Using an example, describe the *resource pooling* principle in the context of network interfaces, and how this concept improves on multihoming.

[2 marks]

iii. Describe in detail the goals of the MultiPath TCP (MPTCP) coupled congestion control algorithm.

[4 marks]

iv. Discuss how a mobile device connected to the Internet using multiple access network technologies will be affected by its use of MPTCP.

[4 marks]

Total 25 marks

Question 3

Software Defined Networking (SDN) and Networking Functions Virtualisation (NFV) are both relatively new concepts within the networking industry.

3.a Discuss the changing landscape under which modern networks are now operating. Include in your discussion the requirements that modern networks need to address, and why classical approaches to computer networking are potentially struggling to realise this vision.

[6 marks]

3.b Explain how SDN is designed to address these concerns.

[4 marks]

3.c OpenFlow is commonly used in SDN networks. Explain the difference between *Pure OpenFlow* and *Hybrid OpenFlow* and discuss the associated trade-offs of using each approach.

[4 marks]

3.d Explain the core concepts behind NFV, and discuss why this would be particularly attractive to ISPs and network operators.

[6 marks]

3.e You have been commissioned by an ISP to investigate the use of NFV within their business. Using a use case scenario of your own choosing, highlight how the scenario would differ as a result of NFV concepts being applied. Your answer should include two diagrams, one illustrating the current scenario using a traditional networking approach and another showing the potential scenario with NFV.

[5 marks]

Total 25 marks

Question 4

4.a A Content Distribution Network (CDN) is an infrastructure designed to efficiently deliver content to the users. One of the challenges within CDNs is selecting the "best" CDN node to provide content to clients. Suggest two ways in which this selection might be approached.

[2 marks]

4.b Explain why HTTP is an appropriate transport technology for Internet video delivery.

[4 marks]

4.c Explain the role of the Media Presentation Description (MPD) in MPEG-DASH playback.

[3 marks]

4.d Explain and relate the following terms in the context of the current public internet architecture:

i. Tier 1 ISPs
ii. Peer ISPs
iii. Internet Exchange Points
iv. CDNs **[4 marks]**

4.e Compare and contrast Transit vs Settlement Free Peering. **[2 marks]**

4.f Imagine you are a network designer for a small consumer broadband access ISP.

i. In choosing how to route your consumer traffic what two types of factor are you most concerned about?

[2 marks]

ii. Based on these two factors, discuss the merits of connecting and routing your subscriber traffic directly to:

a) other small ISPs,
b) an internet exchange point,
c) content delivery networks (CDNs), or
d) a large transit provider? **[8 marks]**

Total 25 marks

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